

ALEXIS MARCHAND

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EMPLOYMENT

- | | |
|--------------------------|--|
| from 9/2025
to ... | IM PAN, Warszawa (Poland)
Adiunkt (Assistant Professor)
Research project: <i>Analysis on groups</i> (NCN Maestro 13)
Mentor: PIOTR NOWAK |
| from 5/2026
to 7/2026 | Universität Regensburg / Humboldt-Stiftung (Germany)
Humboldt Research Fellow
Research project: <i>Invariants of groups and non-positive curvature</i>
Mentor: CLARA LÖH |
| from 9/2024
to 9/2025 | Kyōto University / JSPS (Japan)
JSPS Postdoctoral Fellow
Research project: <i>Stable commutator length in finitely presented groups</i>
Mentor: KŌJI FUJIWARA |

EDUCATION

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|----------------------------|---|
| from 10/2021
to 10/2024 | University of Cambridge (U.K.) • <i>Ph.D. in Pure Mathematics</i>
Thesis: <i>Computations and lower bounds for scl and the relative Gromov seminorm</i>
Defended on: 15/7/2024
Funding: EPSRC PhD studentship
Supervisor: HENRY WILTON |
| from 9/2017
to 8/2021 | École Normale Supérieure de Lyon (France) • <i>Dipl. ENS</i>
Funding: ENS studentship awarded by national competitive examination (rank: 16) |
| from 9/2020
to 7/2021 | Université Grenoble Alpes (France) • <i>M.Sc. in Mathematics</i>
Mention Très Bien
Thesis: <i>Automorphismes extérieurs de produits libres : revêtements abéliens caractéristiques et représentations libres</i>
Supervisor: FRANÇOIS DAHMANI |
| from 10/2019
to 6/2020 | University of Cambridge (U.K.) • <i>M.A.St. in Mathematics</i>
Pass with Merit
Essay: <i>Rigidity theorems for hyperbolic groups</i>
Supervisor: HENRY WILTON |
| from 9/2017
to 7/2019 | ÉNS de Lyon / Université Lyon 1 (France) • <i>B.Sc. in Mathematics</i>
First two years of ENS curriculum in mathematics (L3+M1) |

from 9/2015 | **Lycée Louis le Grand, Paris** (France) • *Preparatory class*
to 7/2017 | Mathematics, physics, computer science

from 9/2012 | **Lycée Gustave Flaubert, Rouen** (France) • *Baccalauréat*
to 7/2015 | Scientific pathway, British international option

PUBLICATIONS (PEER-REVIEWED)

- [1] Alexis Marchand, From letter-quasimorphisms to angle structures and spectral gaps for scl. **Israel J. Math.** (online first). DOI:[10.1007/s11856-026-2898-3](https://doi.org/10.1007/s11856-026-2898-3) • arXiv:[2402.13856](https://arxiv.org/abs/2402.13856)
- [2] Alexis Marchand, Bavard duality for the relative Gromov seminorm, **Enseign. Math.** **72** (2026), no. 1-2, 1-46. DOI:[10.4171/lem/1087](https://doi.org/10.4171/lem/1087) • arXiv:[2305.02149](https://arxiv.org/abs/2305.02149)
- [3] Alexis Marchand, Isometric embeddings of surfaces for scl, **Groups Geom. Dyn.** **19** (2025), no. 4, 1445-1478. DOI:[10.4171/ggd/845](https://doi.org/10.4171/ggd/845) • arXiv:[2302.04133](https://arxiv.org/abs/2302.04133)
- [4] Rémi Coulon and Alexis Marchand, Growth and co-growth in negatively curved groups, in Dahmani, Erschler, Horbez, Wise (eds.), Groups acting on fractals, hyperbolicity and auto-similarity, **Panoramas et Synthèses** **63** (2025).
- [5] Alexis Marchand, Free representations of outer automorphism groups of free products via characteristic abelian coverings, **J. Group Theory** **26** (2023), no. 2, 399-420. DOI:[10.1515/jgth-2021-0154](https://doi.org/10.1515/jgth-2021-0154) • arXiv:[2107.11230](https://arxiv.org/abs/2107.11230)

EXPOSITORY ARTICLES

- [6] Alexis Marchand, Makers of patterns: from Escher to Coxeter, **Eureka** (2020).
- [7] Alexis Marchand, Géométrie des groupes et courbure négative, **Journal de Mathématiques des Élèves de l'ENS de Lyon** (2020).

GRANTS & FUNDING

from 9/2024 | **JSPS** • Fellow's Research Support Allowance
to 9/2025 | Allocated amount: JPY 840,000
Topic: *Stable commutator length in finitely presented groups*

6/2025 | **RIMS, Kyōto University** • RIMS Workshop (type A)
Allocated amount: JPY 350,000
Topic: *Spectrum problems for scl via S-machines*

AWARDS

3/2023 | **Smith-Knight & Rayleigh-Knight prize** • University of Cambridge
Grade 3 (GBP 300), for the essay *Isometric embeddings of surfaces for scl*

RESEARCH VISITS

11/2025 | **INI, Cambridge** (U.K.) • Three-week visit
Programme: *Operators, graphs, groups* (participant)

6/2025	RIMS, Kyōto University (Japan) • Two-week small-group workshop Programme: <i>Spectrum problems for scl via S-machines</i> (co-organiser)
6/2025	KIAS, Seoul (South Korea) • One-week visit • invited by SANG-HYUN KIM
4/2025	University of Hawai'i at Mānoa (U.S.) • Ten-day visit • invited by YASH LODHA
4/2022 and 6/2022	IHP, Paris (France) • One two-week visit and one three-week visit Programme: <i>Groups acting on fractals</i> (participant)
from 6/2019 to 8/2019	IMPA, Rio de Janeiro (Brazil) • Two-month M.Sc. internship Topic: <i>Topology of complex affine varieties</i> • Supervisor: HOSSEIN MOVASATI
from 5/2018 to 6/2018	KIT, Karlsruhe (Germany) • Six-week B.Sc. internship Topic: <i>Geometry of Coxeter groups</i> • Supervisor: CATERINA CAMPAGNOLO

HIGHER EDUCATION TEACHING EXPERIENCE

from 1/2023 to 5/2024	Geometry • Trinity College & Peterhouse College, Cambridge Part IB supervisions • 34 contact hours • in English
from 1/2024 to 3/2024	Geometric Group Theory • University of Cambridge Part III examples classes • 8 contact hours • in English
from 10/2022 to 1/2024	Analysis & Topology • Trinity College & Peterhouse College, Cambridge Part IB supervisions • 46 contact hours • in English
from 12/2022 to 12/2023	Admissions interviews • Christ's College, Cambridge Co-interviewer for undergraduate admissions in mathematics • in English
from 1/2023 to 5/2023	Group Cohomology • University of Cambridge Part III drop-in sessions • 4 contact hours • in English
from 1/2022 to 5/2022	Riemann Surfaces • Trinity College & Peterhouse College, Cambridge Part II supervisions • 31 contact hours • in English
from 10/2021 to 1/2022	Linear Algebra • Trinity College, Cambridge Part IB supervisions • 16 contact hours • in English
from 10/2021 to 1/2022	Mapping Class Groups • University of Cambridge Part III drop-in sessions • 4 contact hours • in English
from 9/2018 to 5/2019	Colles de Mathématiques • Lycée Aux Lazaristes, Lyon Oral examinations in mathematics for first-year students in a preparatory class • 44 contact hours • in French

VOLUNTEER TEACHING

3/2025	JSPS science dialogue • Koshi High School, Fukui Lecture in English and Japanese for a high school science club
from 10/2023 to 4/2024	High school project supervision • Rusanivskyi Lyceum, Kyiv (online) Student: OLHA MYKHAILYK • Topic: <i>Inversions and Möbius transformations</i> Second prize at the city stage in the Junior Academy of Sciences of Ukraine research competition

from 10/2019 to 3/2020	Tutoring • Coleridge Community College & St Faith's School, Cambridge Volunteer tutoring in mathematics in secondary schools, under the <i>Stimulus</i> programme
from 9/2017 to 6/2020	French as a foreign language • ItinérENS, Lyon French classes for refugees • more than 100 contact hours

SERVICE

Refereeing for *Trans. Amer. Math. Soc.*, *Algebr. Geom. Topol.*, *KIAS Springer Ser. Math.*

from 9/2025 to ...	Analytic & geometric group theory seminar • IM PAN, Warszawa Co-organiser
9/2025	World of GroupCraft V • Online Co-organiser
6/2025	RIMS Workshop • RIMS, Kyōto University Topic: <i>Spectrum problems for scl via S-machines</i> • Co-organiser
from 2/2025 to 4/2025	Reading seminar • Kyōto University Topic: L^2 -Betti numbers • Organiser
from 9/2023 to 8/2024	Junior geometry seminar • University of Cambridge Co-organiser
from 8/2023 to 5/2024	MCR committee • Trinity College, Cambridge MCR (Middle Combination Room) committee member in charge of: the MCR's website, mailing list, elections, lunchtime seminars, etc.
from 10/2022 to 12/2022	Reading seminar • University of Cambridge Topic: <i>Generalisations of hyperbolic groups</i> • Organiser

INVITED TALKS

6/2026	ICMAT, Madrid Advanced school: Aspherical 2-complexes and their fundamental groups: past, present, and future
6/2026	Université Bourgogne Europe Séminaire géométrie, algèbre, dynamique et topologie
6/2026	Université de Lorraine Séminaire de géométrie différentielle
5/2026	Universität Regensburg LKS-Seminar (Oberseminar für Gruppen und geometrische Topologie)
11/2025	Newcastle University Algebra & geometry seminar
11/2025	University of Cambridge Geometric group theory seminar
9/2025	IM PAN, Warszawa Analytic & geometric group theory seminar
7/2025	University of Tōkyō Tuesday seminar on topology

7/2025	Tōhoku University Geometry seminar
6/2025	KIAS, Seoul Topology seminar
5/2025	Kyōto University Differential topology seminar
4/2025	University of Hawai'i at Mānoa Mathematics department colloquium
4/2025	Kanazawa University Topology seminar
11/2024	Kyōto University First MMS workshop for young researchers
11/2024	Online Séminaire virtuel francophone groupes et géométrie
5/2024	Università di Pisa / Scuola Normale Superiore Seminari dei baby geometri
4/2024	Online International young seminar on bounded cohomology & simplicial volume
4/2024	Purdue University (online) Geometry and geometric analysis seminar
3/2024	University of Oxford Geometric group theory seminar
12/2023	University of Bristol Bristol junior geometry seminar
6/2023	Online International young seminar on bounded cohomology & simplicial volume
4/2023	KU Leuven Methusalem junior seminar
3/2023	ETH Zürich Geometry graduate colloquium
2/2023	University of Oxford Junior topology & group theory seminar
2/2023	University of Cambridge Geometric group theory seminar
6/2022	Université Grenoble Alpes Séminaire compréhensible
4/2022	Aix-Marseille Université Séminaire des doctorants de St-Charles

CONFERENCES ATTENDED

1/2026	Uniwersytet Jagielloński, Kraków Perspectives in ergodic theory, randomness, and geometry
11/2025	INI, Cambridge Stability and probabilistic methods

2/2025	University of Tōkyō 20th East Asian conference on geometric topology • Contributed talk
1/2025	VAST, Hanoi Geometry and dynamics in low dimensions • Contributed talk
4/2024	UPV / EHU, Bilbao Groups, topology, & applications Gran Bilbao
3/2024	Universität Bielefeld Topological & homological methods in group theory
2/2024	CIRM, Marseille GAGTA • Contributed talk & poster
7/2023	Heriot-Watt University, Edinburgh Postgraduate group theory conference • Contributed talk
5/2023	CRM, Montréal Geometry of subgroups
1/2023	University of Southampton LMS regional meeting • Contributed poster
9/2022	Universität Regensburg Recent advances in bounded cohomology
6/2022	IHP, Paris Hyperbolic groups and their generalisations
4/2022	IHP, Paris Mapping class groups and $\text{Out}(F_n)$
4/2022	CIRM, Marseille Groups acting on fractals • Note-taker for a minicourse [4]
2/2022	KIT, Karlsruhe Workshop on finiteness properties

LANGUAGES

- French • Native language
- English • Fluent (C2)
Passed the CAE (**Cambridge**)
- Japanese • Advanced (B2)
Passed the JLPT N2
- German • Advanced (B2)
- Portuguese • Advanced (B2)
- Russian • Intermediate (A2/B1)
- Polish • Intermediate (A2/B1)

June 15, 2026